

CARE Hospital Management System

Authors

Mohamed Ibrahim	231002063
Ibrahim Ahmed	231001317
Tasnim Maged	231000173
Mohamed Shalaby	231000546
Ahmed Taher	231001502

Description

This is a management system made with C programming language which is used to keep the hospital data (CARE Hospital). The project manipulates some files that keep these data. There are four basic files for the project: a file for available rooms, another one for patients' records, another one for available slots for appointments, and the last one for booked appointments. First, the program asks the user (The person who has access to the management system) to choose what they want.

Details

The user will choose a number referring to the procedure that will be done.

Case (1)

If the user chooses 1, They can insert new records to the hospital files. They can insert either a patient (with a room), or someone with an appointment (the user will specify). If someone is inserted, an available place will be removed from the "available rooms" file or the "available slots" file. The user is supposed to enter their name, credit card number, and CVV to finish the payment and billing procedure. Mohamed Ibrahim and Ibrahim Ahmed did this part.

Case (2)

If the user chooses 2, they can remove either a patient's record or a booked appointment record (the user will specify). If a record is removed, an available place will be added to the "patients" file or the "appointments" file. Mohamed Shalaby and Tasnim Maged did this part

Case (3)

If the user chooses 3, they will choose what to display on the screen. They can display patients' records, booked appointments, available rooms, and available slots. The data will be read from the files. Ahmed Taher did this part.

Case (4)

In the case of 4, "About us" window will be printed. It is read from a file. Mohamed Ibrahim did this part.

Case (5)

In the case of 5, the program will terminate and leave a message "The program has terminated...".

In all cases, the user will be asked if they want to go back to the home page or not. If they choose "yes", the program will be repeated. Tasnim Maged did this part.

Test Cases

1) Book an appointment

Name: Ahmed Muhammad

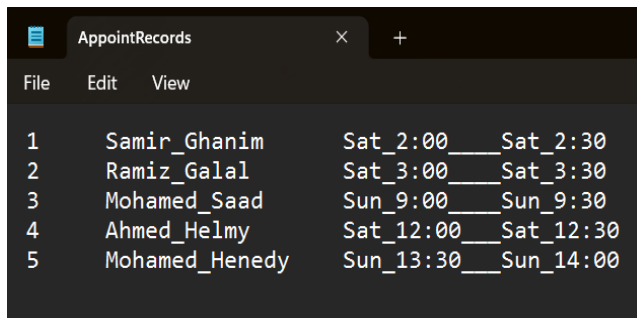
Credit card number: 1234-5678-9012-3456, CVV: 123

Slot number: 6

Results:

Before:

AppointRecords.txt

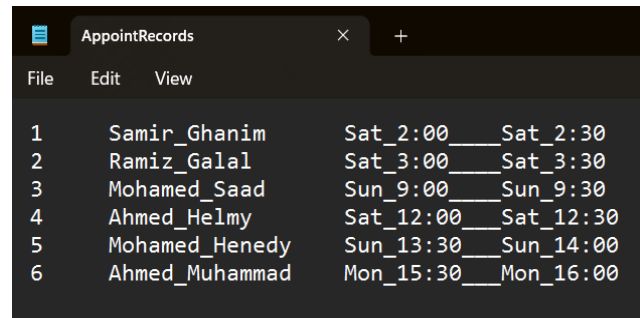


A screenshot of a text editor window titled 'AppointRecords' showing a list of 5 appointments. Each entry consists of an ID, a name, and a date-time range.

1	Samir_Ghanim	Sat_2:00__Sat_2:30
2	Ramiz_Galal	Sat_3:00__Sat_3:30
3	Mohamed_Saad	Sun_9:00__Sun_9:30
4	Ahmed_Helmy	Sat_12:00__Sat_12:30
5	Mohamed_Henedy	Sun_13:30__Sun_14:00

After

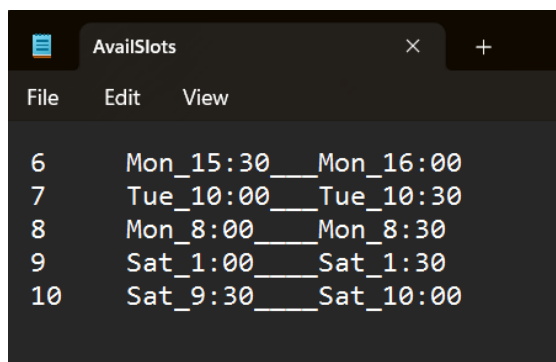
AppointRecords.txt



A screenshot of a text editor window titled 'AppointRecords' showing a list of 6 appointments. The 6th entry, 'Ahmed_Muhammad Mon_15:30__Mon_16:00', has been added to the previous list.

1	Samir_Ghanim	Sat_2:00__Sat_2:30
2	Ramiz_Galal	Sat_3:00__Sat_3:30
3	Mohamed_Saad	Sun_9:00__Sun_9:30
4	Ahmed_Helmy	Sat_12:00__Sat_12:30
5	Mohamed_Henedy	Sun_13:30__Sun_14:00
6	Ahmed_Muhammad	Mon_15:30__Mon_16:00

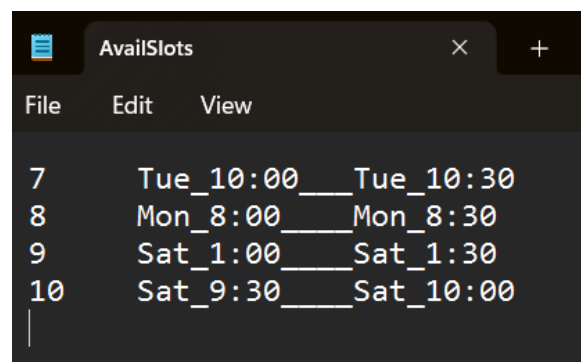
AvailSlots.txt



A screenshot of a text editor window titled 'AvailSlots' showing a list of 5 available slots. Each entry consists of an ID and a date-time range.

6	Mon_15:30__Mon_16:00
7	Tue_10:00__Tue_10:30
8	Mon_8:00__Mon_8:30
9	Sat_1:00__Sat_1:30
10	Sat_9:30__Sat_10:00

AvailSlots.txt



A screenshot of a text editor window titled 'AvailSlots' showing a list of 6 available slots. The 6th entry, 'Tue_10:00__Tue_10:30', has been added to the previous list.

7	Tue_10:00__Tue_10:30
8	Mon_8:00__Mon_8:30
9	Sat_1:00__Sat_1:30
10	Sat_9:30__Sat_10:00

2) Inserting a patient

Name: Ahmed Muhammad

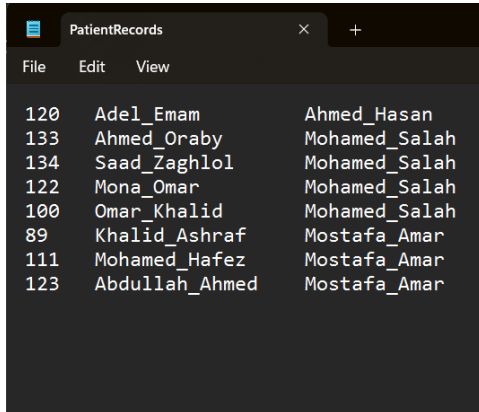
Credit card number: 1234-5678-9012-3456

CVV: 123

Results:

Before:

PatientRecords.txt

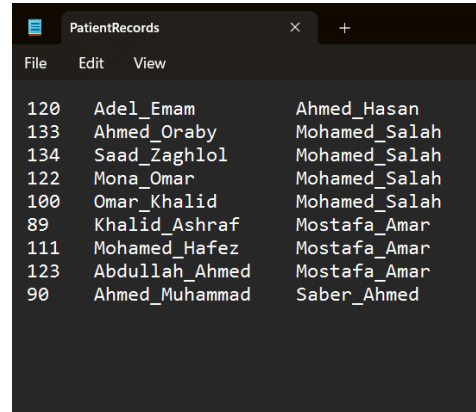


A screenshot of a text editor window titled 'PatientRecords' showing a list of patient records. The records are displayed in three columns: ID, Name, and Doctor. The list includes 8 entries with IDs ranging from 89 to 134.

120	Adel_Emam	Ahmed_Hasan
133	Ahmed_Oraby	Mohamed_Salah
134	Saad_Zaghlo1	Mohamed_Salah
122	Mona_Omar	Mohamed_Salah
100	Omar_Khalid	Mohamed_Salah
89	Khalid_Ashraf	Mostafa_Amar
111	Mohamed_Hafez	Mostafa_Amar
123	Abdullah_Ahmed	Mostafa_Amar

After

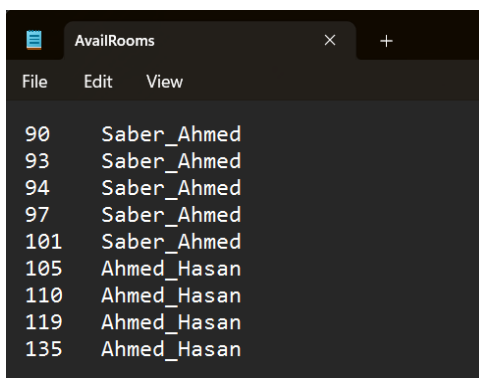
PatientRecords.txt



A screenshot of a text editor window titled 'PatientRecords' showing the updated list of patient records. The records are displayed in three columns: ID, Name, and Doctor. The list includes 9 entries, with the addition of a new record at the bottom (ID 90) and a change in the ID of the last record (from 123 to 90).

120	Adel_Emam	Ahmed_Hasan
133	Ahmed_Oraby	Mohamed_Salah
134	Saad_Zaghlo1	Mohamed_Salah
122	Mona_Omar	Mohamed_Salah
100	Omar_Khalid	Mohamed_Salah
89	Khalid_Ashraf	Mostafa_Amar
111	Mohamed_Hafez	Mostafa_Amar
123	Abdullah_Ahmed	Mostafa_Amar
90	Ahmed_Muhammad	Saber_Ahmed

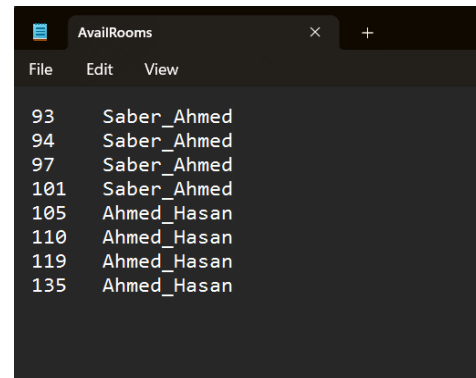
AvailRooms.txt



A screenshot of a text editor window titled 'AvailRooms' showing a list of available rooms. The rooms are displayed in two columns: ID and Name. The list includes 8 entries with IDs ranging from 90 to 135.

90	Saber_Ahmed
93	Saber_Ahmed
94	Saber_Ahmed
97	Saber_Ahmed
101	Saber_Ahmed
105	Ahmed_Hasan
110	Ahmed_Hasan
119	Ahmed_Hasan
135	Ahmed_Hasan

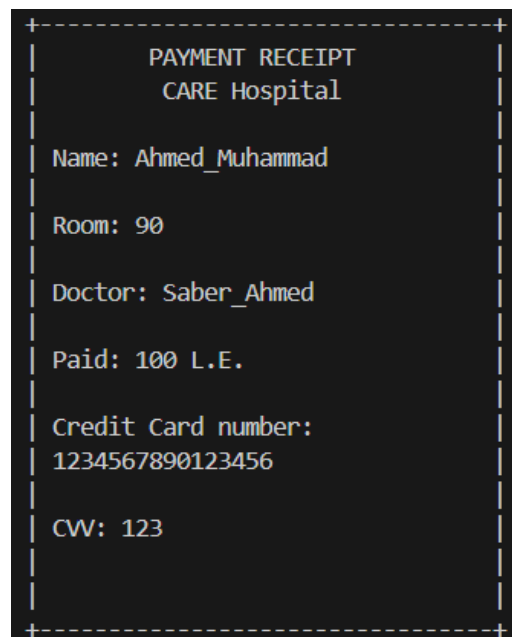
AvailRooms.txt



A screenshot of a text editor window titled 'AvailRooms' showing the updated list of available rooms. The rooms are displayed in two columns: ID and Name. The list includes 8 entries, with the addition of a new room at the top (ID 93) and a change in the ID of the last room (from 135 to 101).

93	Saber_Ahmed
94	Saber_Ahmed
97	Saber_Ahmed
101	Saber_Ahmed
105	Ahmed_Hasan
110	Ahmed_Hasan
119	Ahmed_Hasan
135	Ahmed_Hasan

Bill



A screenshot of a payment receipt form titled 'PAYMENT RECEIPT CARE Hospital'. The form is enclosed in a dashed border and contains the following information:

NAME: Ahmed_Muhammad

Room: 90

Doctor: Saber_Ahmed

Paid: 100 L.E.

Credit Card number:
1234567890123456

CW: 123

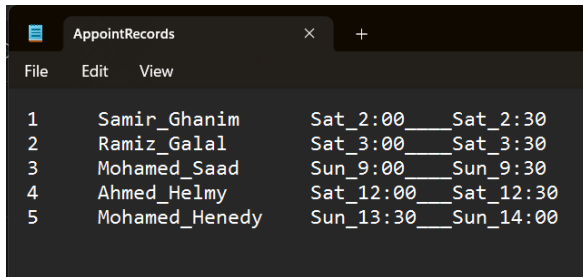
3) Removing a booked appointment

Removing the appointment at slot 4:

Results:

Before

AppointRecords.txt

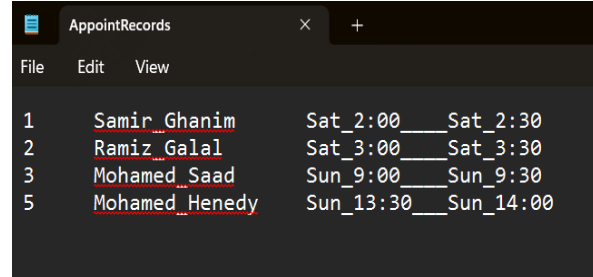


A screenshot of a text editor window titled 'AppointRecords' showing a list of 5 appointments. Each line contains an ID, a name, and a time slot. The names are underlined.

1	Samir_Ghanim	Sat_2:00__Sat_2:30
2	Ramiz_Galal	Sat_3:00__Sat_3:30
3	Mohamed_Saad	Sun_9:00__Sun_9:30
4	Ahmed_Helmy	Sat_12:00__Sat_12:30
5	Mohamed_Henedy	Sun_13:30__Sun_14:00

After

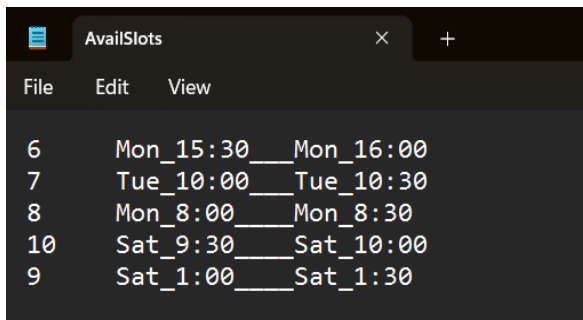
AppointRecords.txt



A screenshot of a text editor window titled 'AppointRecords' showing a list of 5 appointments. The appointment at slot 4 has been removed. The names are underlined.

1	Samir_Ghanim	Sat_2:00__Sat_2:30
2	Ramiz_Galal	Sat_3:00__Sat_3:30
3	Mohamed_Saad	Sun_9:00__Sun_9:30
5	Mohamed_Henedy	Sun_13:30__Sun_14:00

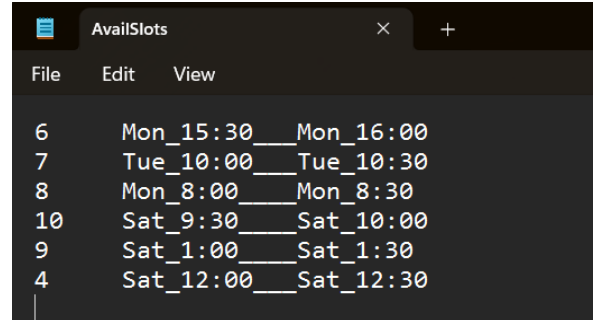
AvailSlots.txt



A screenshot of a text editor window titled 'AvailSlots' showing a list of 9 available slots. Each line contains an ID, a start time, and an end time.

6	Mon_15:30__Mon_16:00
7	Tue_10:00__Tue_10:30
8	Mon_8:00__Mon_8:30
10	Sat_9:30__Sat_10:00
9	Sat_1:00__Sat_1:30

AvailSlots.txt



A screenshot of a text editor window titled 'AvailSlots' showing a list of 9 available slots. The slot at index 4 has been removed. The order of the remaining slots is preserved.

6	Mon_15:30__Mon_16:00
7	Tue_10:00__Tue_10:30
8	Mon_8:00__Mon_8:30
10	Sat_9:30__Sat_10:00
9	Sat_1:00__Sat_1:30
4	Sat_12:00__Sat_12:30

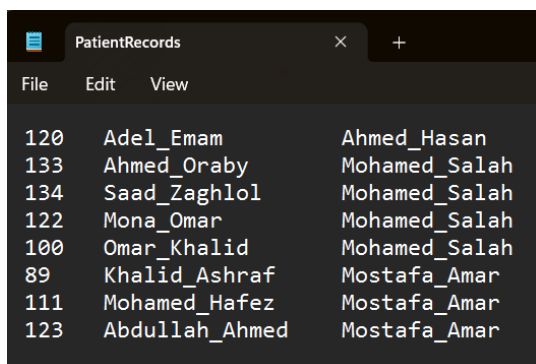
4) Removing a patient record

Removing the patient at room 89

Results:

Before:

PatientRecords.txt

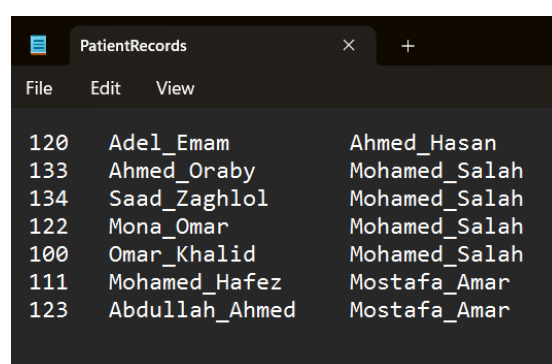


A screenshot of a text editor window titled 'PatientRecords' showing a list of 10 patients. Each line contains an ID, a name, and a room number. The names are underlined.

120	Adel_Emam	Ahmed_Hasan
133	Ahmed_Oraby	Mohamed_Salah
134	Saad_Zaghlo1	Mohamed_Salah
122	Mona_Omar	Mohamed_Salah
100	Omar_Khalid	Mohamed_Salah
89	Khalid_Ashraf	Mostafa_Amar
111	Mohamed_Hafez	Mostafa_Amar
123	Abdullah_Ahmed	Mostafa_Amar

After:

PatientRecords.txt



A screenshot of a text editor window titled 'PatientRecords' showing a list of 9 patients. The patient at room 89 has been removed. The order of the remaining patients is preserved.

120	Adel_Emam	Ahmed_Hasan
133	Ahmed_Oraby	Mohamed_Salah
134	Saad_Zaghlo1	Mohamed_Salah
122	Mona_Omar	Mohamed_Salah
100	Omar_Khalid	Mohamed_Salah
111	Mohamed_Hafez	Mostafa_Amar
123	Abdullah_Ahmed	Mostafa_Amar

AvailRooms.txt

File	Edit	View
90	Saber_Ahmed	
93	Saber_Ahmed	
94	Saber_Ahmed	
97	Saber_Ahmed	
101	Saber_Ahmed	
105	Ahmed_Hasan	
110	Ahmed_Hasan	
119	Ahmed_Hasan	
135	Ahmed_Hasan	

AvailRooms.txt

File	Edit	View
90	Saber_Ahmed	
93	Saber_Ahmed	
94	Saber_Ahmed	
97	Saber_Ahmed	
101	Saber_Ahmed	
105	Ahmed_Hasan	
110	Ahmed_Hasan	
119	Ahmed_Hasan	
135	Ahmed_Hasan	
89	Mostafa_Amar	

5) Showing the data of all patients

Results:

What do you want to display:

- (1) Patients
- (2) Booked appointments
- (3) Available rooms
- (4) Available slots

1

Room	Name	Doctor
120	Adel_Emam	Ahmed_Hasan
133	Ahmed_Oraby	Mohamed_Salah
134	Saad_Zaghlol	Mohamed_Salah
122	Mona_Omar	Mohamed_Salah
100	Omar_Khalid	Mohamed_Salah
89	Khalid_Ashraf	Mostafa_Amar
111	Mohamed_Hafez	Mostafa_Amar
123	Abdullah_Ahmed	Mostafa_Amar